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Experiment 7: Breadth First Search (BFS) in Python

Aim

To write a Python program to implement the **Breadth First Search (BFS)** algorithm.

Algorithm

1. Create a graph using an adjacency list.
2. Start from the source node.
3. Mark the source node as visited.
4. Visit all adjacent unvisited nodes using a queue.
5. Repeat until the queue becomes empty.
6. Print the BFS traversal.

Program

```
from collections import deque

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

visited = set()
queue = deque(['A'])

print("BFS Traversal: ", end="")
```

```
while queue:
    node = queue.popleft()

    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbour in graph[node]:
            if neighbour not in visited:
                queue.append(neighbour)

print("\n\nBFS Tree:")
print("        A")
print("       / \\")
print("      B   C")
print("     / \\   \\")
print("    D   E   F")
```

Input

Graph:
A → B, C
B → D, E
C → F
D → None
E → None
F → None

Starting Node = A

Output

```
BFS Traversal: A B C D E F

BFS Tree:
        A
       / \
      B   C
     / \   \
    D   E   F
```

Result

Thus, the Python program to implement the Breadth First Search (BFS) algorithm was executed successfully, and the BFS traversal of the graph was displayed.